CO1

1) **Write a python program multiply 2 arrays of the same size element by element.**
2) **Program to create a vector with values from 0 to 20 and change the sign.**
3) **Program to check whether two arrays are equal (element wise) or not**
4) **Program to create a 3X3 identity matrix**
5) **Program to create a 5x5 zero matrix with elements on the main diagonal.**
6) **Draw a line in a diagram from position (1,3) to (2,10) then to (6,12) and finally to position (18,20). (Mark each point with a red color and line style is dotted).**
7) Write a Python program to draw a line using given axis values taken from a text file, with suitable label in the x axis, y axis and a title.
8) **Draw a plot for the for the following data:**

   Temp: 12, 14, 16, 18, 20, 22, 24

   Celsius: 100, 200, 250, 400, 300, 450, 500

9) Create a dictionary and convert it in to corresponding dataframe.

10) Write a python program to Generate the series of dates from 1st May, 2021 to 12th May, 2021 (both inclusive).

11) Given a 2D List, convert it into corresponding dataframe and display it.

12) **Given a CSV file, read it into a dataframe and display it.**

13) **Given a dataframe, sort it by multiple columns.**

CO2

1)**Program to implement k-NN classification using any standard dataset available in the public domain and find the accuracy of the algorithm.**

2) **Program to implement Naïve Bayes Algorithm using any standard dataset available in the public domain and find the accuracy of the algorithm.**

3)**Program to implement linear and multiple regression techniques using any standard dataset available in the public domain and evaluate its performance.**

CO3

1)**Program to implement text classification using Support vector machine**

2) **Program to implement decision trees using any standard dataset available in the public domain and find the accuracy of the algorithm.**

3) Program to implement k-means clustering technique using any standard dataset available in the public domain.